

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q17-0842

Burrows Crossing Road 138 kV
August 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0842. The Interconnection Customer proposes a 231.9 MW natural gas combined cycle facility interconnecting at the Burrows Crossing Road 138 kV substation in the Virginia area. The request was received on August 03, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 249 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,465,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Falu 1334 – Burrows Crossing Rd Substation 138 KV Line, the Burrows Crossing Rd – Falu 1334 138 KV Line 1 loads to 103.4% of its 440 MVA rating. Project DFAX: 34.3%.

Van Dom – Hayfield 138 KV Line: 102.3% loading (286 MVA) following the outage of the Burnt Church – Falu 1334 138 KV line; DFAX = 22.8%.

For the loss of Falu 1334 – Belvoir 138 KV Circuit 2, the Belvoir – Falu 1334 138 KV loads to 101.5% of its 263 MVA rating. Project DFAX: 13.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Falu 1334 – Burrows Crossing Rd 138 KV #1	Reconductor line	\$1,001,000	78% (\$784,000)
Van Dom – Hayfield 138 KV Line	Reconductor line	\$517,000	72% (\$373,000)
Belvoir – Falu 1334 138 KV	Reconductor line	\$1,308,000	100% (\$1,308,000)
	Total allocated to project		\$2,465,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.